

## Lectures 7-8 DSP + Machine Learning Instructor Package

### Audience

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Senior undergraduate + first-year graduate ECE students.

### Texts

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- Oppenheim & Schaffer, Discrete-Time Signal Processing, Chapter 7 focus.  
- Raschka, Machine Learning with PyTorch and Scikit-Learn, CNN / PyTorch connections.

### Lecture 7 – FIR Filter Design, Windowing, & 1D Convolutions

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Approx. 30 slides.

Teaching arc:

1. FIR vs. IIR from the phase / ML perspective.
2. Linear phase, symmetry, and group delay.
3. Ideal lowpass derivation using the inverse DTFT.
4. Rectangular truncation and Gibbs phenomenon.
5. Window method and transition-width / sidelobe tradeoff.
6. Rectangular, Bartlett, Hann, Hamming, Blackman comparison.
7. FIR convolution as a fixed-weight PyTorch Conv1d operation.
8. Cross-correlation convention in Conv1d and tap reversal for textbook convolution.
9. Fixed DSP taps vs. learned convolution kernels.
10. Homework 1-4 + instructor solution appendix.

### Lecture 8 – IIR Design, Bilinear Transformation, & Data Preprocessing

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Approx. 37 slides.

Teaching arc:

1. Butterworth, Chebyshev I/II, and elliptic analog prototypes.
2. Butterworth magnitude-squared response and order intuition.
3. Bilinear transform and stability mapping.
4. Frequency warping derivation.
5. Prewarping and a worked  $0.4\pi$  example.
6. Explicit SciPy analog-prototype -> bilinear design path.
7. IIR nonlinear phase and group delay.
8. Forward-backward zero-phase filtering for offline ML preprocessing.
9. Why filtfilt-style processing squares magnitude, is noncausal, and can create train/deployment skew.
10. Homework 5-8 + full weekly quiz + solutions.

### Code

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lecture7\_fir\_windowing\_conv1d.py

Generates FIR/window figures and verifies fixed PyTorch Conv1d against textbook convolution.

lecture8\_iir\_bilinear\_preprocessing.py

Generates analog-prototype, bilinear/prewarp, phase, and forward-backward filtering figures.

notebooks/

Student-facing notebooks plus executed verification copies for both lectures.

Homework and quiz

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Lectures7\_8\_Homework\_Quiz\_Solutions.md contains 8 homework problems with complete worked solutions and an 8-question weekly quiz with solutions.

PowerPoint compatibility

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The \*\_Microsoft\_PowerPoint.pptx versions were round-tripped through python-pptx and successfully rendered after creation. Equations are embedded as LaTeX-rendered images for reliable display.

Code verification

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- Lecture 7 PyTorch/NumPy convolution mismatch: approximately  $3.6e-7$  in float32.
- Lecture 8 prewarped 4th-order Butterworth design: measured -3 dB cutoff approximately  $0.3999\pi$  for target  $0.4\pi$ .
- Executed notebooks contain no execution errors.